1) Attribute provides additional information about an HTML element. Attributes are written inside opening tag.

Google

<a> - is anchor tag. href - is an attribute

2) CSS layout is the process of arranging HTML elements on webpage. It controls position of elements. width & height. spacing, alignment & responsiveness.

Different CSS layout methods:

1. Normal Flow:
   This is the default layout of HTML elements
   Block elements appear one below another.
2. Flexbox:
   Flexbox is used to arrange elements in a row or column.
3. Grid
   Grid is used to create layouts with rows & columns.
4. positioning:
   positing controls where an element appears.
5. Responsive layout:
   Responsive design makes a webpage adapt to different screen sizes.

3). A programming language is a language used to communicate instructions to a computer.

We use programming languages to build websites, Develop mobile apps, create games, process data & Automate tasks.

4. Javascript is a programming language primarily used to make websites interactive & dynamic

Features:

1. javascript allows variables to hold different types of values
2. javascript generally executes one piece of javascript code at a time on a given thread.
3. javascript supports to events
4. javascript supports callback, promises, async await

5. A function is a reusable block of code that performs a specific tasks.

Types:

- Function Declaration
- function expression
- Arrow function
- Anonymous function
- callback function.
- Higher order function

6. The fetch API is a build-in browser API used to send HTTP requests & recieve responses from servers

How it works:

1. Send HTTP request to API.
2. Recive HTTP response.
3. parse JSON response body
4. use the received data in javascript.

7. React is a javascript library used to build use interfaces. It was developed by meta. React helps developers build reusable UI components & manage changes to the interface

Features

1. Component based architecture
2. Reusability
3. Declarative UI
4. Virtual DOM
5. State management.

8. ES stands for Ecma script. It is the standard specification for javascript.

ES6 also known as ES2015 introduced many important features to java script.

ES6+ refers to ES6 & later versions of ECMA script.
Several features where introduced.

9. npm stands for Node package manager.

It is used to install package, manage, dependices, Run project scripts & update packages

npx stands for Node package executor

Used to execute package commands, runs executable packages, runs package executables.

10. Babel is a javascript compiler. It transforms modern javascript syntax into code that can run in environments with different levels of language support used to transform jsx into js.

- Esbuild is a fast javascript bundler & transformer.
supports bundling, javascript transformation, jsx transformation, minification. It is known for high speed.

- webpack is module bundler. It takes application modules & dependencies & bundles them into output files.
Webpack supports javascript, CSS, image, plugins, loads, code splitting.

- A Bundler takes different files & modules used by an application & combines or organizes them into output files that can be used by the browser

11. Vite is a modern frontend development tool that provides a development server & production build.

- It is commonly used with React, Vue, javascript & typescript.

features of vite

1. Fast development server
2. Hot module Replacement
3. production build
4. supports modern frontend tools
5. Easy setup.

12. OXC :

Oxc is a collection of high performance javascript tool written in Rust.

It includes tools for parsing, linting, Transformation, minification

- Rollup is a javascript module bundler. It bundles javascript modules into optimized output.
Rollup is known for tree shaking, ES module support, plugin ecosystem, Library bundling.
- Rolldown is rust based javascript bundler designed to be compatible with Rollups ecosystem. It aims to provide faster bundling while supporting rollup setup configuration & plugins.
- Vite has been adopting Rolldown based tooling for its bundling pipeline.

13. npm run dev - used to start the development server.

purpose: develop the application, provides changes.
Use hot module replacement.

npm start - Runs the script named start.

- A development server, A backend server, A production server

npm run build - used to create production ready output

purpose: compile code, bundle files, optimize assets
prepare files for deployment.

14. JSX stands for javascript XML.

It is a syntax extension commonly used with React to describe UI using HTML like syntax inside javascript.

Difference:

jsx:	javascript
Syntax extension	programming language
used to describe UI	used for logic & behavior
Looks similar to HTML	uses JS syntax.
commonly use with React	used in many environment.

15.